



Predicting Hospital Readmissions for Diabetic Patients

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PROBLEM STATEMENT



- Why is hospital readmission prediction important?
 - Unplanned hospital readmissions lead to **higher healthcare costs** and **poorer patient outcomes**.
 - Identifying high-risk patients can **improve care** and **optimize hospital resources**.
- **Goal:** Develop a machine learning model to **predict patient readmission** and assist in early intervention.

Dataset Overview



Source: Kaggle's *Diabetes 130-US hospitals* dataset



Number of Records: 100,000+ patient admissions



Features: 50 patient attributes (demographics, medications, lab tests, etc.)



Target Variable: 'Readmitted' (<30, >30, NO)



Challenges: - Imbalanced, Messy and Incomplete data requiring intensive data cleaning and transformation.



Features Description

Demographics: Age, gender, race

Admission Details: Admission type, discharge disposition, admission source

Medical History: Number of diagnoses, medication changes

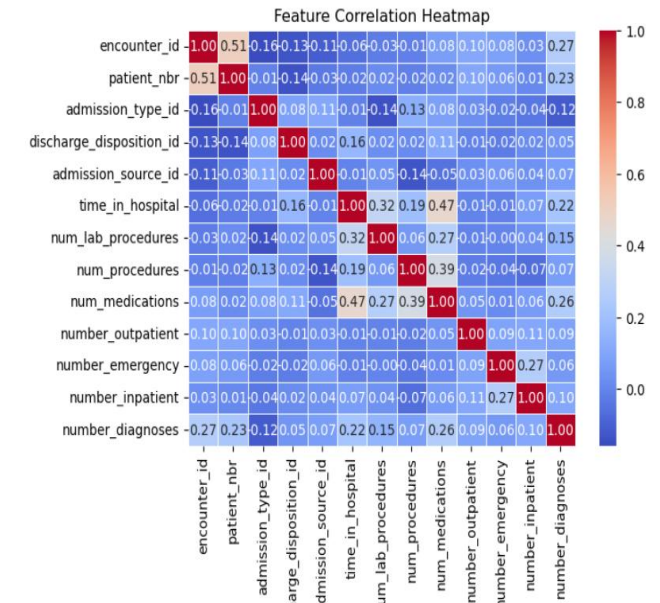
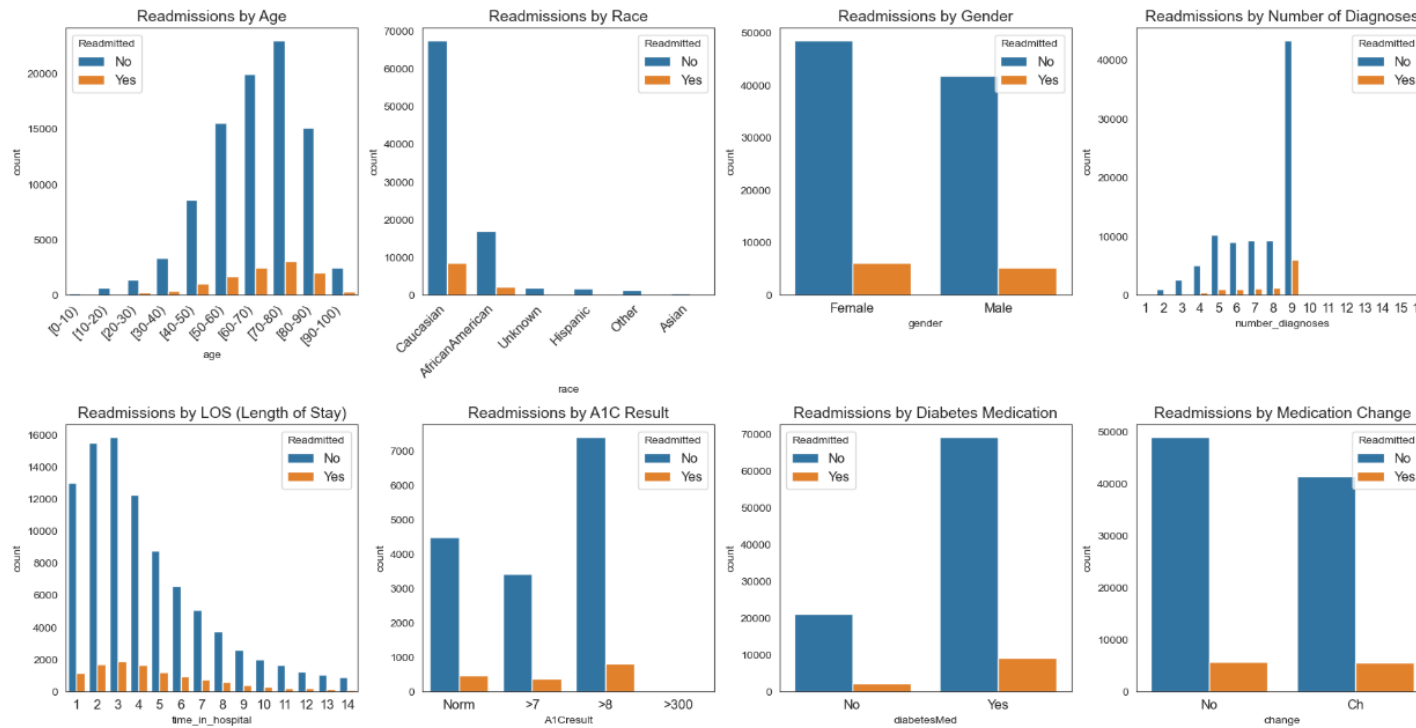
Lab Results: Glucose serum levels, A1C test results

Medications: 23 drugs prescribed, including insulin

Hospital Stay Info: Time in hospital, number of procedures, number of medications

Exploratory Data Analysis

Readmissions by Feature



Data Cleaning & Preprocessing

Handling Missing Values:

weight	98569
medical_specialty	49949
payer_code	40256
race	2273
diag_3	1423
diag_2	358
diag_1	21

- Dropped columns with excessive nulls

(weight:97%, medical_specialty:49%,
payer_code:29%)

- Imputed 'race' with an 'Unknown' category
- Removed data for gender with
'Unknown/Invalid' category

Encoding Categorical Features:

- 'gender', 'change' and 'diabetesMed' :
Binary Encoded
- 'discharge_disposition_id',
'admission_source_id' : Label Encoded(high
cardinality)
- 'race', 'max_glu_serum', 'A1Cresult' : One-
Hot Encoded

Feature Engineering & Transformations

Feature Engineering:

ICD-9 Code Categorization:

- Diagnoses in 'diag_1', 'diag_2' and 'diag_3' were mapped into broader **disease categories** (Circulatory, Respiratory, Digestive, Diabetes, etc.).

Aggregated Prior Visits:

- The sum of **inpatient, outpatient, and emergency visits** into a new feature 'num_visits'

Log Transformations:

Highly skewed variables (**num_medications, num_visits**) were log-transformed for better model performance.

Categorical Mappings (Admissions, Discharge, ICD-9 Codes)

During preprocessing, several categorical features were **mapped into meaningful categories** to improve model performance.

Admission Source ID Mapping:

- Encoded different admission sources (e.g., emergency, referral, transfer).
- Helps in identifying **risk levels based on admission type**.

Discharge Disposition ID Mapping:

- Removed patients **discharged to hospice or deceased** to avoid skewed predictions.
- Encoded various discharge types for meaningful classification.

ICD-9 Diagnosis Codes Mapping:

Grouped over 1000 diagnosis codes into 9 broader medical categories:

- Circulatory Diseases (1)
- Respiratory Diseases (2)
- Digestive System
- Diabetes
- Injury
- Musculoskeletal Disorders
- Genitourinary
- Neoplasms (Cancer) (8)
- Other Unclassified (0)

Feature Selection

- Reduce redundant features
- Avoids Overfitting
- Improve model performance
- RFE (Recursive Feature Elimination) with XGBoost was used to select the most important features.
- The final selected features included demographics, admission details, medication use, and diagnostic categories.

- **Selected Features:**

```
['age', 'admission_type_id', 'discharge_disposition_id', 'admission_source_id', 'time_in_hospital',  
'num_procedures', 'number_diagnoses', 'diabetesMed', 'ICDCat1', 'ICDCat2', 'ICDCat3',  
'number_changes', 'number_meds', 'race_Caucasian', 'max_glu_serum_>300',  
'max_glu_serum_None', 'A1Cresult_None', 'A1Cresult_Norm', 'num_visits_log',  
'num_medications_log']
```

Machine Learning Approach

Step-1: Addressing Class Imbalance

Step-2: Feature Scaling (Standardization)

Step-3: Train-test split (80-20)

Step-4: Model tuning using **GridSearchCV**

Step-5: Cross-validation (Stratified K-Fold)

Step-6: Evaluation metrics → Accuracy, Precision, Recall, F1-score, AUC-ROC

Class Imbalance

Techniques Used:

- SMOTE (Synthetic Minority Over-sampling Technique)
- Class Weighting Balancing
- Balanced Bagging

Findings:

- SMOTE improved recall but affected precision.

	Accuracy	Precision	Recall	F1-Score	AUC-ROC
LR	0.89	0.00	0.00	0.00	0.6400
LR (balanced class_weight)	0.63	0.17	0.56	0.26	0.6400

Logistic Regression

- Feature scaling was applied to standardize continuous variables.
- Imbalance handling significantly improved recall (from **0.00 to 0.55**).
- Without handling imbalance, the model predicted **only the majority class**.

	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Baseline model	0.63	0.17	0.56	0.26	0.64
Tuned	0.63	0.17	0.56	0.26	0.64
CV results	0.62	0.16	0.55	0.25	0.63

Decision Tree Model

- Baseline DT model had poor recall (0.15)
- **Tuning improved recall significantly to 0.61**, but overall performance remained **moderate**.
- Decision Trees **tend to overfit**, which is why **ensemble methods (RF & XGB) were explored**.
- **Decision Trees** are easy to interpret but **prone to overfitting**.
- 'max_depth': 5, 'min_samples_leaf': 5, 'min_samples_split': 2

	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Baseline model	0.81	0.15	0.15	0.15	0.52
Tuned	0.57	0.16	0.64	0.25	0.64
CV results	0.57	0.15	0.64	0.25	0.63

Random Forest Model

- Poor recall, imbalanced class issue.
- Using SMOTE improved recall but slightly reduced precision.
- Tuned Model: Adjusted using n_estimators, max_depth, min_samples_split
- **Key finding:** RF was better than DT but still suffered from **class imbalance**.
- 'max_depth': 10, 'min_samples_split': 10, 'n_estimators': 300

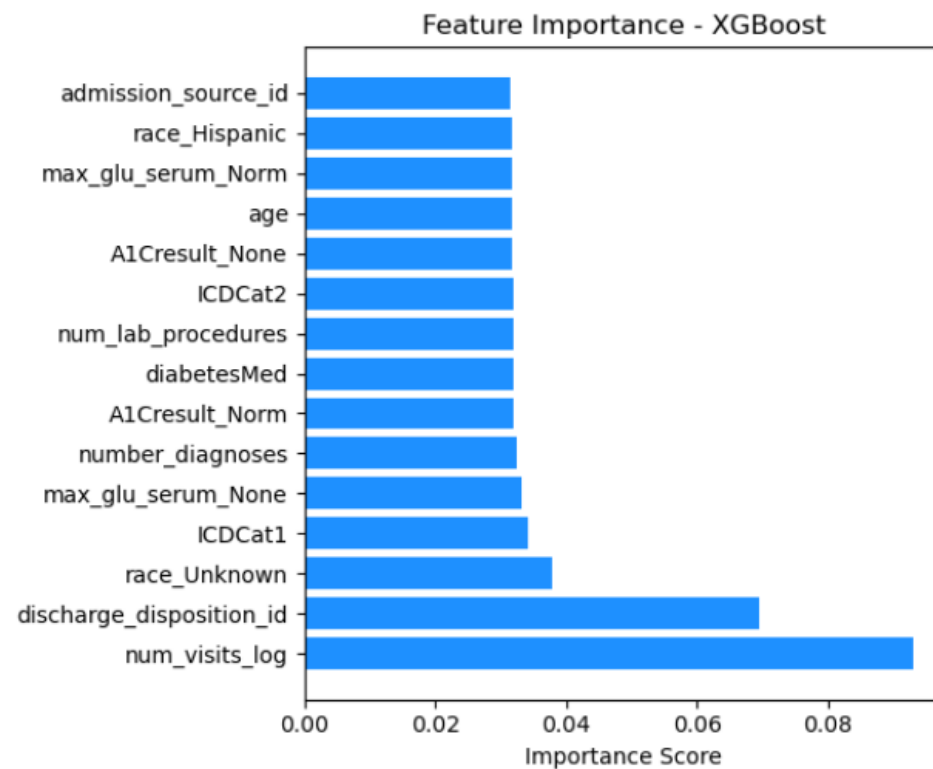
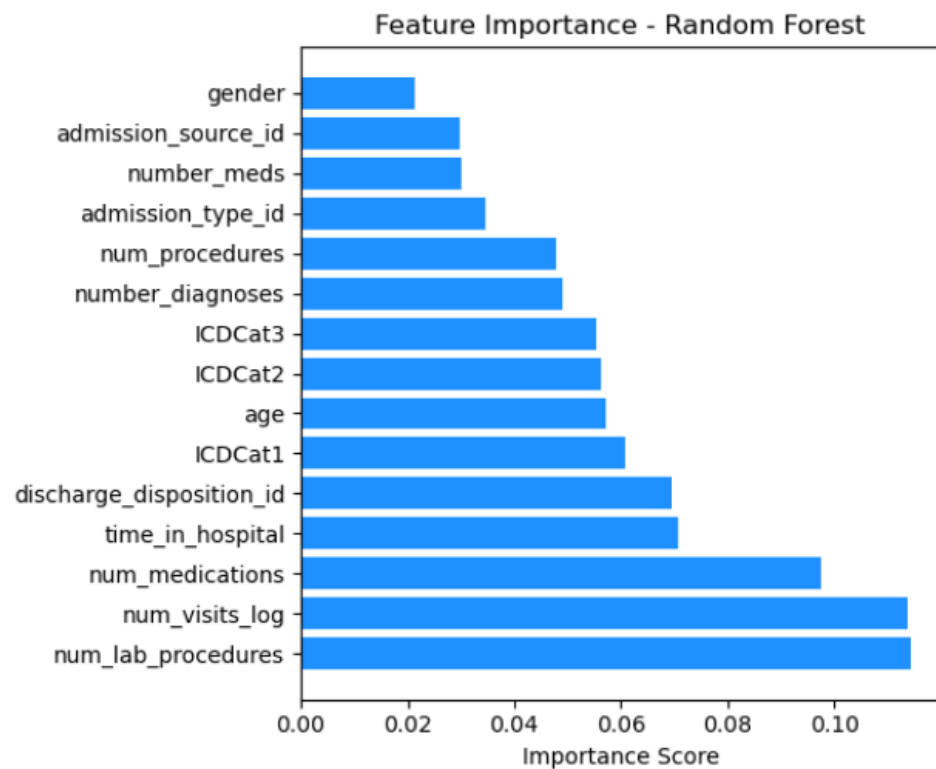
	Accuracy	Precision	Recall	F1-Score	AUC-ROC
RF	0.89	0.50	0.00	0.10	0.62
RF(SMOTE)	0.83	0.19	0.14	0.16	0.61
RF Tuned	0.66	0.18	0.55	0.27	0.65
CV results	0.67	0.17	0.51	0.26	0.65

XGBoost Model

- **Why XGBoost?** Handles missing data, feature importance ranking.
- **Best Parameters:** max_depth=10, n_estimators=300, min_samples=10
- XGBoost outperformed other models in recall
- Tuning further boosted recall from 0.39 to 0.43
- Best model achieved AUC-ROC = 0.67

	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Base model	0.59	0.16	0.63	0.26	0.65
Gridsearch	0.48	0.15	0.78	0.25	0.66
Tuned Model	0.74	0.20	0.42	0.27	0.66
CV Results	0.75	0.19	0.38	0.26	0.65

Feature Importance Analysis



Ensemble Models (Voting Classifier)

- An ensemble model was trained using Decision Trees, Random Forests and XGBoost for enhanced generalization.
- Resulted in the best performance overall with 43% recall, 20% precision and 73% accuracy.
- Ensemble **improved recall and F1-score without overfitting.**

	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Balanced Bagging	0.65	0.18	0.58	0.27	0.66
Ensemble (DT+RF+XGB)	0.73	0.20	0.43	0.27	0.66

Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.63	0.17	0.56	0.26	0.64
Decision Trees	0.57	0.16	0.64	0.25	0.64
Random Forests	0.66	0.18	0.55	0.27	0.65
Gradient Boosting	0.74	0.20	0.42	0.27	0.66
Ensemble (Voting)	0.73	0.20	0.43	0.27	0.66
Balanced Bagging	0.65	0.18	0.58	0.27	0.66

Key Takeaways & Insights

- XGBoost performed **best individually**, but **ensemble models improved further**.
- Readmission is influenced by diagnosis, visits, medications, and hospital stay.
- Class imbalance required special handling (**SMOTE, class weights, ensemble**).

Business Impact:

- Early identification of high-risk patients → **Better resource allocation**.
- Reducing readmissions → **Lower hospital costs & better patient care**.
- Hospitals can **use predictive models** to optimize treatment plans.

Limitations & Future Work

Limitations:

- Imbalanced dataset → Despite tuning, precision for minority class remained low.
- Feature set could be expanded to individual medications.
- PCA could have been done

Future Work:

- Compare with other models like SVMs and KNN.
- Implement **Deep Learning models (Neural Networks, LSTMs)**.